



A composite transportation network design problem with land-air coordinated operations[☆]

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ABSTRACT

With the advent of electric vertical-takeoff-and-landing (eVTOL) vehicles, it becomes imperative to assess their impact on transportation network efficiency. Hence, this paper introduces a novel model for the composite transportation network design problem (CTNDP). Specifically, the model is designed to inform the planning of composite transportation networks by considering travel behaviors of land-air coordinated mobility. This innovative model is formulated as a bi-level programming problem, in which the upper-level model focuses on strategically planning vertiport locations and capacities, thereby reducing the total travel time within the transportation system. Upon determining decisions at the upper level, the composite transportation network can be established by constructing new links that interconnect the vertiports and existing road networks. The lower-level model addresses the land-air collaboration network equilibrium (LAC-NE) problem through mathematical programming. To solve the bi-level programming problem, a customized Bayesian optimization method, namely the mixed-integer Bayesian optimization (MI-BO) method, is proposed. Specifically, MI-BO is developed by incorporating the branch-and-bound algorithm into the Bayesian optimization framework. Additionally, an advanced path-based improved gradient projection (IGP) algorithm is developed to efficiently resolve the LAC-NE problem. The efficacy of the proposed model and solution algorithms are substantiated through numerical experiments, and the results illustrate the impact of land-air coordinated operations on the performance of the composite transportation network.

1. Introduction

Transportation is a vital facilitator of human activities, and with a growing population, the demand for efficient transportation is escalating. The resulting surge in road traffic contributes to more and more mobility issues like congestion, air pollution, and accidents; particularly in bustling city centers. Addressing these challenges necessitates strategic transport network planning and traffic control strategies. These concerns are conventionally tackled within the urban transportation network design problem (UTNDP) (Dantzig et al. 1979; Yang and Wang 2002; Cantarella et al. 2006; Farahani et al. 2013; Gu et al. 2018; Alkaabneh et al. 2019; Jiang and Nielsen 2022;

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The advent of electric vertical-takeoff-and-landing (eVTOL) vehicles is reshaping transportation systems and introducing novel mobility options. Different from traditional helicopters, this innovative mode offers distinct advantages, including reduced noise, heightened safety, cost-effectiveness, and enhanced environmental sustainability. These qualities position eVTOL vehicles as key players in the future of urban transportation. Notably, both high-tech companies and aviation industries are actively developing business models utilizing eVTOL vehicles for daily commuting, fostering the concept of urban aerial mobility (UAM). Globally, countries including China, Singapore, the United States, and Japan, along with organizations such as the European Union Aviation Safety Agency, Joby Aviation, and EHang, are making substantial investments in UAM infrastructure and technologies. These developments are critical to the integration of eVTOL vehicles into daily commuting. Given the swift evolution of UAM technologies, it is expected that UAM will soon become a widely adopted mode of urban transportation systems. With the growth of eVTOL vehicles, a key concern is understanding the role that the UAM plays in an integrated transportation system and how it can effectively coordinate with traditional road transportation.

It is worth noting that drones have recently been employed for parcel delivery, effectively shifting urban logistics into the airspace (Bombelli et al. 2020; Tseremoglou et al. 2022). In light of this evolving landscape, it is increasingly clear that eVTOL vehicles will be pivotal in passenger transportation, propelling urban aerial mobility into an inevitable future. Notably, due to the unique properties of eVTOL vehicles, integrating them directly into existing road networks presents challenges, as this could potentially disrupt the regular traffic flow of conventional vehicles. This highlights the need to address traffic assignment challenges within a three-dimensional space. A crucial strategy involves establishing several vertiports, serving as virtual nodes strategically positioned to alleviate these concerns. Building on this, it becomes evident that the transition to UAM necessitates the development of new transportation infrastructures designed to accommodate this innovative mode. Among the critical factors, the strategic placement and capacity configuration of vertiports are crucial, significantly influencing traveler behavior. Moreover, empowering sustainable mobility through advanced analytics and optimization is crucial in shaping the future of transportation into a green, flexible, and well-synchronized mobility ecosystem that meets the needs of society. However, note that existing studies on UTNDP primarily focus on traditional ground-based modes of transport, such as automobiles, subways, and buses—within a two-dimensional framework, often overlooking the integration of advanced low-altitude (ALT) transportation. To realize a green, efficient, and seamless experience for travelers, it is essential to optimize transportation planning and operations by incorporating this emerging transportation mode. Therefore, this paper focuses on addressing the composite transportation network design problem with land-air coordinated operations, aiming to provide a comprehensive analytical framework for its modeling and solution algorithm.

1.1. Literature review

The UTNDP has been under consistent investigation for the past six decades, with a growing number of publications over time (Dantzig et al. 1979; Yang and Wang 2002; Mesbah et al. 2008; Scarinci et al. 2017; Pu et al. 2022; Kim et al. 2023). This can be attributed to the problem's inherent complexity, theoretical intrigue, and practical significance. The UTNDP is typically formulated as a bi-level problem. The upper-level problem pertains to the decision-maker, often a government authority, responsible for planning or managing the transport network. This problem involves practical policy discussions, measurable goals (e.g., reducing total travel time), and design decisions (e.g., construction of new roads). The assumption is that the decision-maker can predict travelers' travel behaviors. In contrast, the lower-level problem is the traveler's problem, where individuals decide whether to travel and if so, select travel modes and routes based on decisions made at the upper level. The bi-level structure allows the decision-maker to make transportation planning or management decisions by considering traveler reactions and improving network performance.

The upper-level problems of UTNDP can be categorized based on various network design policies and decisions. Traditionally, they are divided into road network design problems and transit network design problems. Among them, the former focuses on street networks without distinguishing between public transit and private vehicles (Wang et al. 2013; Hosseininassab and Shetab-Boushehri 2015; Yin et al. 2017; Di et al. 2018; Wang et al. 2020a; Mansourianfar et al. 2021; Chai et al. 2023; Kim et al. 2023), while the latter considers public transit network topology, service frequency, and timetables (Mesbah et al. 2008; Gallo et al. 2011; Szeto and Jiang 2014; Yu et al. 2015; Iliopoulou and Kepaptsoglou 2021; Lee et al. 2022). Notably, these types of problems mentioned above typically focus on a single mode (e.g., car, metro, or transit) and frequently confine themselves to a single transportation network. In practice, however, multiple modes coexist, and travel demands within different modes are interconnected. A more fitting classification for this scenario is the multi-modal network design problem, representing a category that incorporates at least two distinct modes within the UTNDP framework (Yao et al. 2012; Rashidi et al. 2016; Huang et al. 2018; Pinto et al. 2020; Du et al. 2022).

The lower-level problem takes different forms based on considered choice dimensions and traveler assumptions. Road network design problems allocate traffic flows to road networks based on predefined transport policies. For transit network design problems, the lower-level problem focuses on assigning passengers to transit vehicles, taking into account a predetermined flow of transit vehicles resulting from the specific transit design scenario. Furthermore, the lower-level model is generally employed to describe route choices of travelers, which is known as trip assignment problems, encompassing traffic assignment problems (LeBlanc et al. 1975; Yao et al. 2019; Zhang et al. 2023b) and transit assignment problems (Wahba 2004; Ryu et al. 2017; Xu et al. 2020). Notably, trip assignment assumes route choice behavior principles, such as Wardrop's principle (Wardrop 1952), representing traveler non-cooperation and selection of the shortest path, resulting in user equilibrium (UE) pattern (Beckmann et al. 1956; Bell and Cassir 2002; Guo et al. 2013; Khani 2019; Ma et al. 2022). The extension includes stochastic user equilibrium (SUE) assignment for perceived travel time-based route choice (Daganzo and Sheffi 1977; Xie and Liu 2014; Gu et al. 2022). Opposite to the UE assumption, cooperative traveler behavior leads to system optimal (SO) assignment (Wie et al. 1995; Chow 2009; Rambha et al. 2018).

In what follows, we summarize the problems studied and solution methods used in existing literature, as shown in Table 1. It becomes evident that most UTNDP studies assume fixed demand, with some considering elastic demand. Additionally, a majority focus on a single objective (denoted by S), while a limited number explore multi-objectives (denoted by M). In terms of traffic assignment and modes, the UE principle has received considerable attention. Compared with discrete and mixed design decisions, continuous design decisions attract significant UTNDP research due to the flexibility of variables allowing various modeling approaches and solution methodologies.

In terms of solution methods, the genetic algorithm (GA) has been used in existing literature to tackle mixed-integer decision variables within UTNDP. However, the performance of GA is usually not guaranteed. Recently, the surrogate-based optimization methods have received increasing attention. Surrogate-based optimization fundamentally involves fitting surrogate models—these are cost-effective approximations of complex objective functions—by strategically evaluating the objective function at selected points, known as samples. These models help efficiently pinpoint the optimum's location, significantly reducing the need to compute objective functions directly (Hong and Zhang 2021). Among the various algorithms for surrogate-based optimization, Bayesian optimization (BO) algorithms are prominent for their use of Bayesian statistics in constructing surrogate models (Frazier 2018). BO is noted for its exceptional data efficiency and has been increasingly employed across various transportation issues (Huo et al. 2023; Tay and Osorio 2022; Yin et al. 2022a). This study attempts to develop a BO method for the network design problem. BO is predominantly applied to solve optimization problems with continuous decision variables. However, discrete decision variables widely exist in practical applications. To tackle discrete decision variables, previous studies have adapted the 'naïve approach' for discrete optimization challenges (Garrido-Merchán and Hernández-Lobato 2020; Luong et al. 2019). Specifically, the BO algorithm is directly applied to discrete structures. In this approach, new samples that adhere to integer constraints are created in each iteration by first relaxing these constraints and then rounding to the nearest integer solution. However, this method has been noted in prior studies to potentially result in inefficient sampling (Luong et al. 2019). Some limited studies have explored BO methods for mixed-integer decision variables (Baptista and Poloczek 2018; Cuesta Ramirez et al. 2022). Their focus lies in developing efficient surrogate models to handle mixed-integer decision variables. The sampling method over a feasible set is rarely discussed. This study attempts to develop a tailored BO method for mixed-integer decision variables.

On the other hand, UAM is rapidly gaining momentum, aiming to revolutionize urban transportation by shifting from a two-dimensional to a three-dimensional paradigm. This transformation is largely driven by advancements in eVTOL technology. As these vehicles continue to evolve, UAM is increasingly seen as a promising solution for alleviating ground traffic congestion (Du et al. 2021; Tripathi et al. 2022; Wang et al. 2023; Wang and Qu 2023; Jin et al. 2024; Shon et al. 2024; Yan et al. 2024). However, despite its potential, UAM development faces significant challenges, including infrastructure limitations, regulatory barriers, and safety concerns. Nevertheless, there is growing interest in the planning and operational aspects of UAM, with several studies focusing on path planning for aerial mobility in urban environments (Tang et al. 2021; Pinto Neto et al. 2022). Notably, surveys indicate strong public interest in using UAM for airport shuttle services, highlighting its appeal as a favored mode of transport. Some studies have validated the feasibility of integrating multiple service routes connecting airports using agent-based simulation methods, projecting positive outcomes for the efficiency, safety, and scalability of eVTOL services (Guo et al. 2022; Brunelli et al. 2023b; Lv et al. 2024). Moreover, UAM is expected to operate at high densities and on a large scale, which demands carefully designed operational plans to ensure its safe and efficient implementation. To address this, numerous studies have explored optimization techniques to guide critical decisions in

Table 1
A summary of the studies of UTNDP.

Reference	Single/Multiple Objective	Traffic assignment	Demand	Decision variables	Modes	Solution method
Dantzig et al. (1979)	S	SO	Fixed	Continuous	Car	Lagrangian multiplier
Leblanc and Boyce (1986)	S	UE	Fixed	Continuous	Car	Bard's algorithm
Chen and Alfa (1991)	S	SUE	Fixed	Discrete	Car	Branch and Bound
Yang and Wang (2002)	M	UE	Fixed	Continuous	Car	Simulated annealing
Poorzahedy and Abulghasemi (2005)	S	UE	Fixed	Discrete	Car	Genetic algorithm
Cantarella et al. (2006)	S	UE	Fixed	Mixed	Car	Genetic algorithm
Mesbah et al. (2008)	S	UE	Fixed	Discrete	Car and transit	Enumeration method
Mathew and Sharma (2009)	S	UE	Fixed	Continuous	Car	Genetic algorithm
Long et al. (2010)	S	SUE	Fixed	Discrete	Car	Sensitivity analysis
Gallo et al. (2011)	S	UE	Fixed	Continuous	Car, transit and metro	Genetic algorithm
Hosseiniinasab and Shetab-Boushehri (2015)	S	UE	Fixed	Discrete	Car	Genetic algorithm
Scarinci et al. (2017)	S	UE	Fixed	Continuous	Car	Metaheuristic methods
Li et al. (2019)	S	UE	Fixed	Discrete	Car	Genetic algorithm
Ye et al. 2021	S	UE	Elastic	Continuous	Car, metro, and bike	Successive linear programming method
Wang et al. (2022b)	S	UE	Fixed	Continuous	Car, transit, and bike	Kriging-surrogate-based algorithm
Luan et al. (2023)	S	UE	Fixed	Continuous	Car	Branch and Bound
This paper	S	UE	Fixed	Mixed	Car and eVTOL	Mixed-integer Bayesian optimization

UAM operations, taking into account factors such as energy availability and weather conditions, which significantly impact flight safety (Reiche et al. 2021; Schweiger et al. 2023; Lee et al. 2024).

In addition to vehicle technology and operations, vertiports play a pivotal role in UAM by providing essential infrastructure for take-off, landing, and maintenance. The strategic selection of vertiport locations is crucial for optimizing travel routes, improving service quality, and minimizing urban disruption. Several studies have explored vertiport location optimization at both the network and system levels. For instance, Wu and Zhang (2021) examined the network design problem of UAM on-demand services, focusing on integer programming to determine optimal vertiport locations. Wang et al. (2022a) investigated vertiport placement while considering the interdependencies between strategic deployment, tactical operations, and passenger demand. Macias et al. (2023) developed a linear placement model with vehicle sizing constraints to identify optimal vertiport configurations based on eVTOL performance. Additionally, Brunelli et al. (2023a) conducted a systematic review of vertiport characteristics, identifying key features that influence location and capacity while highlighting research gaps in the field.

Despite these advancements, there remains a need for a more integrated approach to vertiport planning that better examines the relationships between infrastructure, route choice, and travel demand distribution. In this context, our paper focuses on macroscopic aspects, particularly the integration of low-altitude flights with traditional road networks and their impact on commuters' route choices (i.e., the distribution of travel demand). We further investigate how the introduction of UAM influences the overall performance of the transportation system, evaluating its potential to enhance efficiency, reduce congestion, and reshape urban mobility.

1.2. Objectives and contributions

In consideration of the aforementioned points, we contend that the majority of literature addressing the UTNDP primarily concentrates on ground-based transportation modes (e.g., car, transit, and metro). Within this context, numerous efforts have been undertaken to plan an urban transportation network, encompassing strategies like toll setting, parking pricing, and the expansion of existing streets or the construction of new ones. However, crucial distinctions between advanced low-altitude transportation and conventional road transportation underscore the limited applicability of management policies outlined in previous studies. Additionally, in urban environments, the competition for limited land space with other essential facets of city life significantly contributes to persistent and challenging traffic congestion. Thus, the transition from roads to the air assumes particular significance, given the heavy reliance of traditional urban transportation on road infrastructure.

In light of these considerations, we argue that there is an urgent need for the exploration of novel planning and management policies to facilitate seamless land-air coordinated operations, which became especially crucial before the widespread adoption of eVTOL vehicles. In specific, the primary contributions of this paper are threefold. Firstly, we introduce a novel bi-level programming model to formulate the composite transportation network design. In this model, the authority seeks to reduce the total travel time by designing the vertiports' location and capacity (i.e., the upper-level problem). Secondly, based on decisions derived from the authority, the land-air collaboration composite transportation network is established, and commuter route choices within this composite network are characterized following the classical user equilibrium theory, thereby developing a land-air collaboration network equilibrium model. Thirdly, we introduce a tailored algorithm to address the proposed model. Specifically, we have developed a customized mixed-integer Bayesian optimization (MI-BO) method that integrates a modified Gaussian process model and the branch-and-bound technique into the BO framework. This MI-BO approach optimizes the bi-level programming model by strategically selecting feasible solutions, with each solution being assessed through the lower-level problem using the improved gradient projection (IGP) algorithm. Finally, numerical experiments are conducted to validate the applicability of the proposed model, with results offering decision-making support for government authorities.

The subsequent sections of this paper are organized as follows: Section 2 outlines the characterization of a composite transportation network integrating urban aerial mobility. Section 3 delves into the specific network design problem, presenting a bi-level model formulation for planning the land-air coordinated operations. Section 4 provides a customized solution algorithm framework for the proposed model. Section 5 presents numerical examples to demonstrate the effectiveness of the proposed model. Finally, Section 6 concludes this study.

2. The characterization of the land-air collaboration composite network

Given the above, electric vertical-takeoff-and-landing (eVTOL) vehicles are catalyzing the transformation of urban aerial mobility (UAM), offering a more convenient, efficient, and sustainable option for urban commuting. Notably, eVTOL pertains to a unique land-air amphibious vehicle, capable of ground travel akin to conventional vehicles, with the added capability of transitioning to advanced low-altitude flight by accessing the vertiports. Before delving into the specifics of the model, it is essential to provide a summary of the key notations employed in this paper, which are outlined as follows:

Sets:	
N_R	The set of nodes of the road network, and $\forall n \in N_R$;
A_R	The set of links of the road network, and $\forall a \in A_R$;
A_Q	The set of incoming links of the vertiport (termed queue links), and $\forall \bar{a} \in A_Q$;
A_E	The set of outgoing links of the vertiport (termed evacuation links), and $\forall \bar{a} \in A_E$;
A_C	The set of links connecting any two vertiports (termed flight links), and $\forall \bar{a} \in A_C$;

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Sets:	
N_R	The set of nodes of the road network, and $\forall n \in N_R$;
W	The set of OD pairs, and $\forall w \in W$;
K^w	The set of paths between OD pairs w in the land-air collaboration composite network;
P	The set of candidate vertiports, and $\forall p \in P$.
Parameters:	
V_h	The horizontal cruising speed of the eVTOL vehicles;
V_v	The average speed of the vertical take-off and landing of the eVTOL vehicles;
V_f	The free-flow speed on both the road links and the evacuation links;
$L_{\bar{a}}$	The length of the evacuation links, and $\forall \bar{a} \in A_E$;
$L_{\hat{a}}$	The length of the flight links, and $\forall \hat{a} \in A_C$;
h_{lp}	The flight altitude between any two vertiports, and $\forall l, p \in P$;
q^w	The total demand for OD pair w , and $\forall w \in W$;
d	The safe spacing of the eVTOL vehicles during flight;
B	The maximum number of vertiports allowed for construction;
C	The maximum capacity of the constructed vertiports.
Intermediate variables:	
v_a	The traffic flows on the road link a , and $\forall a \in A_R$;
$v_{\bar{a}}$	The traffic flows on the queue link $\bar{a}$, and $\forall \bar{a} \in A_Q$;
$\bar{v}_{\bar{a}}$	The traffic flows on the evacuation link $\bar{a}$, and $\forall \bar{a} \in A_E$;
$v_{\hat{a}}$	The traffic flows on the flight link $\hat{a}$, and $\forall \hat{a} \in A_C$;
t_a	The travel time on the road link a , and $\forall a \in A_R$;
$t_{\bar{a}}$	The travel time on the queue link $\bar{a}$, and $\forall \bar{a} \in A_Q$;
$t_{\bar{a}}$	The travel time on the evacuation link $\bar{a}$, and $\forall \bar{a} \in A_E$;
$t_{\hat{a}}$	The travel time on the flight link $\hat{a}$, and $\forall \hat{a} \in A_C$.
Decision variables:	
$c_{\bar{a}}$	The capacity of the queue links, and $\forall \bar{a} \in A_Q$;
z_p	The binary variables equal 1 when the vertiport is established at the node p , and $\forall p \in P$; otherwise 0;
x_{np}	The binary variables equal 1 when the road node $n \in N_R$ is connected to the vertiport $p \in P$; otherwise 0;
y_{lp}	The binary variables equal 1 when the vertiport $l \in P$ is connected to the vertiport $p \in P$; otherwise 0;
f_k^w	The traffic flow on the route $k \in K^w$ between OD pair $w \in W$.

To enhance a more comprehensive understanding, we present a simplified composite transportation network to illustrate the land-air coordinated operations, as depicted in Fig. 1. Herein, l and p denote the vertiports, serving as the locations where land-air amphibious vehicles execute vertical take-off and landing operations.

Based on the preceding analysis, we posit that the determination of travel time on road links can be modeled using the classical Bureau of Public Road (BPR) function. However, when commuters choose the land-air collaboration travel scheme, as illustrated in Fig. 2, the whole process unfolds in three stages: (i) eVTOL vehicles queue to enter vertiport, execute vertical take-off operation, and reach a safe altitude; (ii) eVTOL vehicles fly to another vertiport near the destination through flight links; (iii) eVTOL vehicles land on the vertiport and then enter road networks via evacuation links. Note that we chose to illustrate a one-way UAM system to demonstrate its integration with traditional road networks, offering an intuitive understanding of UAM operations. However, the subsequent numerical experiments consider a multi-way UAM system, analyzing how the number of vertiports affects total travel time and traffic network performance.

During the first stage, considering vertiport capacity constraints, eVTOL vehicles need to queue to enter vertiports and execute vertical take-off. The travel time for queue links can be characterized based on the classical point queuing theory. Specifically, assuming arrival rates follow a Poisson distribution and service rates for vertical take-off points follow an exponential distribution, the process preceding eVTOL vehicles reaching a safe altitude is akin to an M/M/1/C queuing problem. Consequently, the travel time for queue links is determined by the average stay time in the queuing system, and it can be formulated as (Cho et al. 2020; Liu et al. 2022; Cheng et al. 2022; Zhou et al. 2022; Zhang et al. 2024):

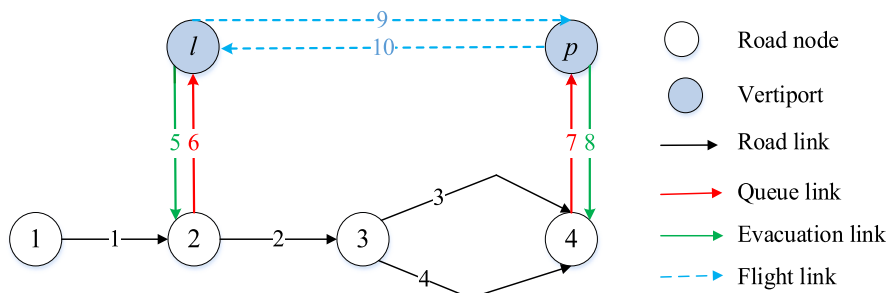


Fig. 1. Diagram of the land-air collaboration composite transportation network.

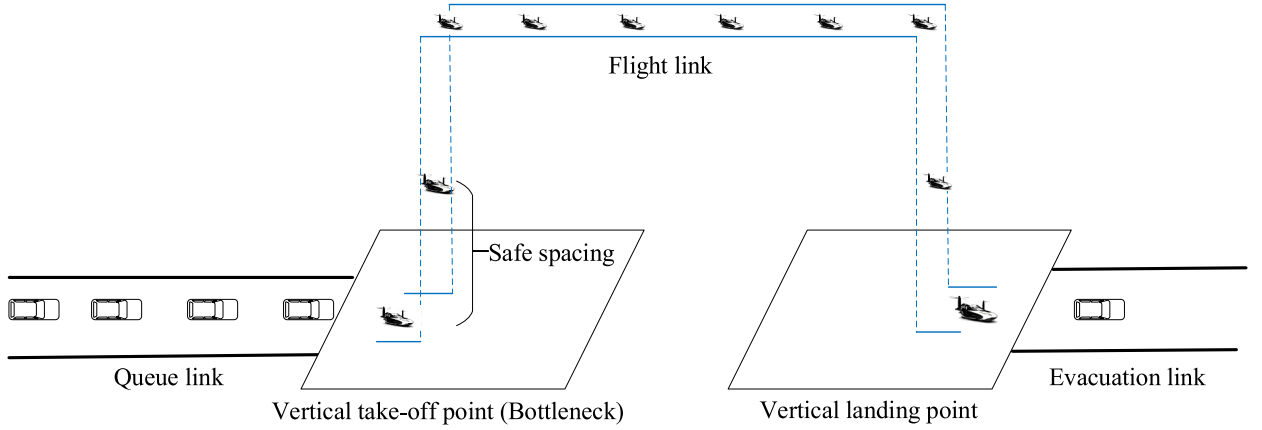


Fig. 2. Schematic diagram of the flight process for eVTOL vehicles.

$$t_{\bar{a}} = \frac{L_{\bar{a}}^q}{\mu(1 - P_{\bar{a}}^0)}, \quad \forall \bar{a} \in A_Q \quad (1)$$

where μ represents the average service rate. Additionally, $L_{\bar{a}}^q$ and $P_{\bar{a}}^0$ indicate the average queue length and the probability of the initial state, respectively, defined as follows:

$$P_{\bar{a}}^0 = \frac{1 - \rho_{\bar{a}}}{1 - (\rho_{\bar{a}})^{c_{\bar{a}}+1}}, \quad \rho_{\bar{a}} \neq 1, \quad \forall \bar{a} \in A_Q \quad (2)$$

$$L_{\bar{a}}^q = \frac{\rho_{\bar{a}}}{1 - \rho_{\bar{a}}} - \frac{(c_{\bar{a}} + 1)(\rho_{\bar{a}})^{c_{\bar{a}}+1}}{1 - (\rho_{\bar{a}})^{c_{\bar{a}}+1}}, \quad \forall \bar{a} \in A_Q \quad (3)$$

where $\rho_{\bar{a}}$ denotes the ratio of the average arrival rate to the average service rate for the specific queue link, and $\rho_{\bar{a}} = \lambda_{\bar{a}}/\mu$. Herein, $\lambda_{\bar{a}}$ denotes the average arrival rate of commuters, which can be determined based on traffic flows corresponding to queue links, namely,

$$\lambda_{\bar{a}} = \sum_{w \in W} \sum_{k \in K^w} f_k^w \chi_{\bar{a},k}^w, \quad \forall \bar{a} \in A_Q \quad (4)$$

where $\chi_{\bar{a},k}^w$ equals 1 when the path k uses the queue link $\bar{a}$ between the OD pair w , and 0 otherwise. Given the average speed of eVTOL vehicles during vertical take-off and landing phases, the average service rate of each vertiport is defined as:

$$\mu = \frac{V_v}{d} \quad (5)$$

where d denotes the safe spacing of the eVTOL vehicles during the flight phase, indicating that when an eVTOL reaches a given safe altitude, the next eVTOL vehicle can be ready for vertical take-off.

In contrast to the first stage, the characterization of travel time is relatively straightforward for both the second and third stages. To ensure the safety and efficiency of eVTOL vehicle operations throughout the flight phase, such as avoiding collisions and hovering over the vertical landing point, it is assumed that eVTOL vehicles maintain a certain cruising speed during the horizontal flight stage, and there is no airspace conflict among various routes. The travel time for flight links comprises two parts: (i) time taken by eVTOL vehicles for vertical take-off and landing (i.e., the first term in Eq. (6)); (ii) time taken by the eVTOL vehicles to fly horizontally (i.e., the second term in Eq. (6)). Thus, the travel time corresponding to flight links is defined as follows:

$$t_{\bar{a}} = \frac{2h_p - d}{V_v} + \frac{L_{\bar{a}}}{V_h}, \quad \forall \bar{a} \in A_C, \quad l, p \in P \quad (6)$$

In this setting, eVTOL vehicles shall sequentially land at the vertiport, effectively avoiding congestion on evacuation links. For simplicity, we assume that eVTOL vehicles travel at free-flow speeds on evacuation links, and hence the corresponding travel time for these links can be calculated as follows:

$$t_{\bar{a}} = \frac{L_{\bar{a}}}{V_f}, \quad \forall \bar{a} \in A_E \quad (7)$$

3. The network planning of land-air coordinated operation

In this section, we elucidate the composite transportation network design problem featuring land-air coordinated operations, followed by the development of an innovative bi-level programming model that integrates the UAM concept.

3.1. The problem description

Building on the insights from Section 2, the characterization of impedance functions for diverse link types (e.g., road links, queue links, flight links, and evacuation links) varies within the land-air collaboration composite transportation network. This approach enables the developed lower-level model not only to simulate traffic flow distribution but also to accurately describe the state of the traffic system under a land-air cooperative operation scenario—a capability that is lacking in traditional traffic assignment models. Furthermore, it is crucial to underscore that both the vertiport locations and the capacities of the queue system play pivotal roles in facilitating the land-air coordinated operations. These factors have a significant impact on the travel time of eVTOL vehicles. Consequently, to fully leverage the advantages of this emerging transportation mode and effectively mitigate urban traffic congestion, there is an urgent need to design the land-air collaboration composite transportation network.

Specifically, we formulate a composite transportation network design problem with land-air coordinated operations as a bi-level programming model (as detailed in Subsection 3.2). This model can be construed as exhibiting the structure of a Stackelberg game between the system policymaker and the network users (Yang and Bell 1998; Yin et al. 2022b). Herein, the objective of the authority (i.e., the upper-level problem) aims to minimize the total travel time (TTT) by determining the vertiport locations and capacities, and the commuters (i.e., the lower-level problem) aim to regulate the flow pattern in the land-air collaboration composite transportation system. The lower-level problem takes the form of a novel land-air collaboration network equilibrium (LAC-NE) model, presented as a mathematical programming problem. In developing this lower-level traffic assignment model, we accounted for the transfer of traffic congestion between ground and low-altitude areas. This consideration is primarily reflected in the characterization of the impedance function for different types of links, which is a key factor influencing the distribution of travel demand.

In general, the upper-level model, representing the authority, is responsible for determining the optimal locations and capacities of vertiports, while the lower-level model, representing the commuters, regulates the distribution of travel demand within the network established by the authority. The upper-level decision on vertiport locations and vertiport capacity shapes the overall topology of the composite network, directly influencing the mode choices and route choices of commuters, as different topologies provide varying sets of feasible paths. Once the upper-level authority determines the vertiport locations and capacities, the lower-level commuters choose their travel paths (i.e., distribute travel demand) based on user equilibrium conditions within the given composite network. The results are then fed back to the upper-level decision-makers, who adjust the vertiport locations and capacities to minimize the total travel time of the transportation system. Commuters subsequently update their travel paths according to the revised network topology, and this iterative process continues until equilibrium is achieved. The bi-level model for planning the land-air collaboration composite network is illustrated in Fig. 3.

3.2. The bi-level model formulation

Given the abovementioned analysis, this section presents the mathematical formulation of the land-air collaboration composite transportation network design problem (LAC-CTNDP), detailed as follows:

[Upper Level].

$$\min f(\mathbf{c}, \mathbf{z}) = \sum_{a \in A_R} v_a t_a + \sum_{a \in A_Q} v_a t_a + \sum_{a \in A_C} v_a t_a + \sum_{a \in A_E} v_a t_a \quad (8)$$

subject to.

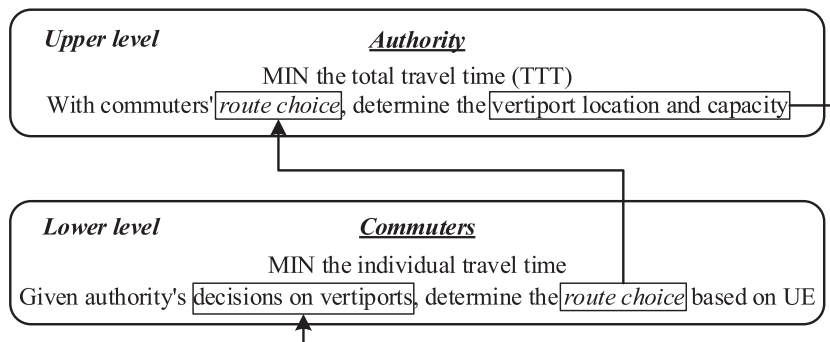


Fig. 3. The bi-level model for planning the land-air collaboration composite transportation network.

$$\sum_{p \in P} z_p \leq B \quad (9)$$

$$c_{\bar{a}} \leq C, \quad \forall \bar{a} \in A_Q \quad (10)$$

$$z_p \in \{0, 1\}, \quad \forall p \in P \quad (11)$$

$$c_{\bar{a}} > 0, \quad \forall \bar{a} \in A_Q \quad (12)$$

[Lower Level].

$$\min Z(\mathbf{f}) = \sum_{a \in A_R} \int_0^{v_a} t_a(w) dw + \sum_{\bar{a} \in A_Q} \int_0^{v_{\bar{a}}} t_{\bar{a}}(w) dw + \sum_{\bar{a} \in A_C} v_{\bar{a}} t_{\bar{a}} + \sum_{\bar{a} \in A_E} v_{\bar{a}} t_{\bar{a}} \quad (13)$$

subject to.

$$\sum_{n \in N_R} X_{np} = z_p, \quad \forall p \in P \quad (14)$$

$$Y_{pl} = z_p z_l, \quad \forall p, l \in P : p \neq l \quad (15)$$

$$v_a = \sum_{w \in W} \sum_{k \in K^w} f_k^w \chi_{a,k}^w, \quad \forall a \in A_R \quad (16)$$

$$v_{\bar{a}} = \sum_{w \in W} \sum_{k \in K^w} \sum_{n \in N_R} \sum_{p \in P} f_k^w \chi_{\bar{a},k}^w X_{np}, \quad \forall \bar{a} \in A_Q \quad (17)$$

$$v_{\bar{a}} = \sum_{w \in W} \sum_{k \in K^w} \sum_{n \in N_R} \sum_{p \in P} f_k^w \chi_{\bar{a},k}^w X_{np}, \quad \forall \bar{a} \in A_E \quad (18)$$

$$v_{\bar{a}} = \sum_{w \in W} \sum_{k \in K^w} \sum_{l \in P} \sum_{p \in P} f_k^w \chi_{\bar{a},k}^w Y_{pl}, \quad \forall \bar{a} \in A_C \quad (19)$$

$$\sum_{k \in K^w} f_k^w = q^w, \quad \forall w \in W \quad (20)$$

$$f_k^w \geq 0, \quad \forall w \in W, k \in K^w \quad (21)$$

$$X_{np} \in \{0, 1\}, \quad \forall n \in N_R, p \in P \quad (22)$$

$$Y_{pl} \in \{0, 1\}, \quad \forall p, l \in P \quad (23)$$

In the upper level, the objective function (8) minimizes the total travel time of the transportation system. Constraint (9) indicates the upper bound on vertiports planned by the authority. Constraints (10) represent the maximum capacity constraints applicable to the queue links (or vertiports). The binary domain of the vertiport location variables is defined in (11). Constraints (12) ensure that the capacity of the queue link remains non-negative.

In the lower level, the objective function (13) is dedicated to minimizing individual travel time within the composite transportation network. Constraints (14) ensure that at least one road node is connected to the constructed vertiports, enabling eVTOL vehicles to enter the vertiports and execute flight operations. Furthermore, to ensure the return of eVTOL vehicles from the vertiports to the road networks, it is crucial to establish a two-way link between the road node and the vertiports. Constraints (15) further guarantee the interoperability of any two constructed vertiports, ensuring that eVTOL vehicles can seamlessly complete low-altitude flights between them. Constraints (16)–(19) define the incidence relationships between path flows and various types of link flows. Constraints (20) ensure flow conservation for travel demand, and the non-negative constraints (21) are imperative for maintaining the physical meaningfulness of the program's solution. Finally, the domains of the binary variables governing the connection between road nodes and vertiports, as well as the connection between vertiports, are defined in (22) and (23).

Challenges for solving the bi-level programming problem mainly lie in the implicitly defined objective function. Specifically, the relationship between the objective function $f(\cdot)$ and the decision variables $\mathbf{c}$ and $\mathbf{z}$ are embedded in the lower-level traffic assignment problem, in which each commuter makes route choices considering the decisions of all other commuters. Consequently, the objective function is highly nonlinear, non-differentiable and intractable. Conventional optimization techniques (e.g., gradient-based methods) are challenging to solve the problem, especially in networks with practical sizes. To tackle this issue, this study proposes a customized Bayesian optimization to inform the design of land-air collaboration composite networks.

4. Solution algorithm

BO is widely used to solve expensive-to-evaluate black-box optimization problems (Frazier 2018). It consists of two iterative steps. First, it constructs a surrogate model using sample points. Second, it suggests and evaluates a new sample for the enrichment of the sample points and the refinement of the surrogate model. These two steps are typically executed by constructing Gaussian process models and optimizing acquisition functions, respectively. This study aims to develop a customized BO method to inform the design of land-air collaboration composite networks. The significant challenge lies in how to handle mixed-integer decision variables. Specifically, two problems need to be addressed. First, the surrogate model should be able to approximate the objective function with mixed-integer inputs. A key component within the Gaussian process model is the kernel function, which measures similarity and covariance between solutions. However, the kernel functions are typically defined for continuous decision variables. The problem lies in how to measure similarity and covariance among solutions when discrete decision variables are included. Second, the acquisition function should be optimized considering integer constraints. Existing literature has shown that the rounding method (i.e., first generating solutions by relaxing all constraints and then rounding to the nearest feasible solution) is inefficient. We aim to explore effective ways to tackle integer constraints. To tackle these problems, a modified Gaussian process model is developed to deal with the mixed-integer decision variables. The modified Gaussian process model is developed by modifying the distance measure in kernel calculation. In addition, regarding the integer restrictions, the branch-and-bound method is incorporated with the commonly used acquisition functions to sample feasible solutions. In BO algorithms, each solution is assessed by solving the lower-level traffic assignment problem. The evaluation can be computationally expensive especially when the networks are of practical scales. To facilitate the evaluation, we incorporate an IGP method into the solution algorithm to efficiently resolve the lower-level LAC-NE problem.

A flowchart summarizing the solution algorithm is depicted in Fig. 4. As illustrated, the MI-BO process unfolds iteratively. Initially, a set of feasible solutions is selected and evaluated at random, creating a dataset that serves as the input for the Gaussian process model tailored for mixed-integer decision variables. This model is used to develop the integer-restricted Expected Improvement (EI) function, a widely adopted acquisition function. A new sample is then generated by optimizing this integer-restricted EI function through the branch-and-bound method. Subsequently, the objective function value of the new sample is assessed. The Improved Gradient Projection (IGP) method is utilized to resolve the lower-level problem. This iterative process continues until the predetermined stopping criteria, which are detailed in Section 4.2, are satisfied.

4.1. Gaussian process for mixed-integer decision variables

Let $\mathbf{x}_i = (\mathbf{z}_i, c_i)$ denote a realization of the decision variable, which is also referred to as a sample point. Suppose we have m ($m \geq 1$) sample points, denoted as $X_m = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$. Given $\mathbf{x}_i \in X_m$, the associated total travel time of the LAC composite transportation network can be obtained by solving the lower-level traffic assignment problem. In this study, we adopted the improved gradient projection (IGP) method to solve the lower-level problem (Xie et al. 2018). Let f_i denote the objective function value associated with variables $\mathbf{x}_i$. Let $F_m = (f_1, \dots, f_m)$ denote the vector of the objective function values. The dataset is defined as $D_m = (X_m, F_m)$. The Gaussian process surrogate model is formulated based on D_m .

A Gaussian process is a set of random variables, any finite number of which have a joint Gaussian distribution. Thus, a Gaussian process $f(\mathbf{x})$ is completely specified by two components: the mean function $g(\mathbf{x})$ and the covariance function $\kappa(\mathbf{x}, \mathbf{x}')$. The two components are defined as:

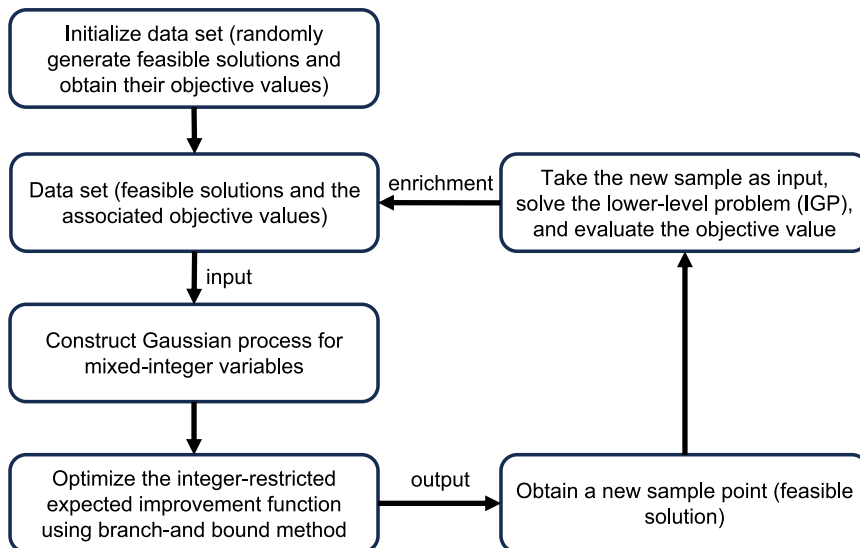


Fig. 4. A flow chart of the MI-BO algorithm.

$$g(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})] \quad (24)$$

$$\kappa(\mathbf{x}, \mathbf{x}') = \mathbb{E}[(f(\mathbf{x}) - g(\mathbf{x}))(f(\mathbf{x}') - g(\mathbf{x}'))] \quad (25)$$

The Gaussian process is written as

$$f(\mathbf{x}) \sim \mathcal{GP}(g(\mathbf{x}), \kappa(\mathbf{x}, \mathbf{x}')) \quad (26)$$

While constructing Gaussian process surrogate models for $f(\cdot)$, a Gaussian process prior is initially applied to $f(\cdot)$, which establishes a joint Gaussian prior distribution. The surrogate model is then derived by conditioning this joint prior distribution on observations at sample points. In the Gaussian process prior, $g(\mathbf{x})$ is typically set to zero. The covariance function is performed using kernel functions. One widely adopted kernel function is the radial basis function (RBF).

$$\kappa(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp(-\|\mathbf{x} - \mathbf{x}'\|_2^2 / 2\eta^2) \quad (27)$$

where σ and η are parameters of the RBF, which can be tuned using maximum likelihood estimation. Eq. (27) explicitly assumes that nearby inputs are highly correlated with outputs. However, the assumption can be violated when dealing with mixed-integer decision variables. In such cases, the covariance or similarity between objective function values cannot be properly captured by the Euclidean distance. To tackle the problem, this study modified the distance measure in kernel calculation. For sample points $\mathbf{x} = (\mathbf{z}, \mathbf{c})$ and $\mathbf{x}' = (\mathbf{z}', \mathbf{c}')$, we propose to use the Hamming distance to measure the similarity in discrete spaces, and the Euclidean distance for continuous spaces.

$$\kappa_z(\mathbf{z}, \mathbf{z}') = \sigma_z^2 \exp\left(-\sum_{i=1}^{d_z} (1 - \delta(z_i, z'_i))\right) / 2\eta_z^2 \quad (28)$$

$$\kappa_c(\mathbf{c}, \mathbf{c}') = \sigma_c^2 \exp(-\|\mathbf{c} - \mathbf{c}'\|_2^2 / 2\eta_c^2) \quad (29)$$

where $\delta(\cdot)$ is the Kronecker delta function. $\delta(z_i, z'_i)$ equals one if $z_i = z'_i$ and zero otherwise. d_z is the dimension of the decision vector $\mathbf{z}$. σ_z (σ_c) and η_z (η_c) are parameters of the kernel functions. Given Eqs. (28) and (29), the kernel function for mixed-integer decision variables is formulated as follows:

$$\begin{aligned} \kappa_{MI}(\mathbf{x}, \mathbf{x}') &= \kappa_z(\mathbf{z}, \mathbf{z}') \kappa_c(\mathbf{c}, \mathbf{c}') \\ &= \sigma_{MI}^2 \exp\left(-\sum_{i=1}^{d_z} (1 - \delta(z_i, z'_i))\right) / 2\eta_z^2 - \|\mathbf{c} - \mathbf{c}'\|_2^2 / 2\eta_c^2 \end{aligned} \quad (30)$$

where $\sigma_{MI} = \sigma_c \sigma_z$. The parameters σ_{MI} , η_z , and η_c can be tuned using maximum likelihood estimation.

Suppose the function $f(\cdot)$ takes the abovementioned Gaussian process prior (i.e., $g(\mathbf{x}) = 0$, and the covariance function takes $\kappa_{MI}(\mathbf{x}, \mathbf{x}')$), the joint prior distribution of $f(\mathbf{x}_1), \dots, f(\mathbf{x}_m)$ and $f(\mathbf{x}')$ is represented in (31), where $\mathbf{x}'$ is the decision vector without evaluation, and we are interested in the associated $f(\mathbf{x}')$.

$$\begin{bmatrix} \mathcal{F}_m \\ f(\mathbf{x}') \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \mathbf{K}_m & \mathbf{K}^* \\ \mathbf{K}^{*T} & \mathbf{K}^{**} \end{bmatrix}\right) \quad (31)$$

where $\mathcal{F}_m = (f(\mathbf{x}_1), \dots, f(\mathbf{x}_m))$, $\mathbf{K}^* = (\kappa_{MI}(\mathbf{x}_1, \mathbf{x}'), \dots, \kappa_{MI}(\mathbf{x}_m, \mathbf{x}'))$, $\mathbf{K}^{**} = \kappa_{MI}(\mathbf{x}', \mathbf{x}')$, and $\mathbf{K}_m$ is an $m \times m$ matrix with entries $\kappa_{MI}(\mathbf{x}_i, \mathbf{x}_j)$, $i, j \in \{1, \dots, m\}$.

Given the observation f_i of $f(\mathbf{x}_i)$, $i \in \{1, \dots, m\}$. The predictive distribution of $f(\mathbf{x}')$ can be derived by conditioning the prior distribution (the distribution (31)) on observations of $f(\mathbf{x}_1), \dots, f(\mathbf{x}_m)$, which yields

$$f(\mathbf{x}') | \mathbf{x}', D_m \sim \mathcal{N}(\mu_f(\mathbf{x}'), \sigma_f^2(\mathbf{x}')) \quad (32)$$

where $\mu_f(\mathbf{x}') = \mathbf{K}^{*T} \mathbf{K}_m^{-1} \mathcal{F}_m$, $\sigma_f^2(\mathbf{x}') = \mathbf{K}^{**} - \mathbf{K}^{*T} \mathbf{K}_m^{-1} \mathbf{K}^*$.

4.2. Branch-and-bound method to solve integer-restricted EI functions

Given the surrogate model of $f(\cdot)$, this section discusses the construction and optimization of acquisition functions, based on which new samples can be selected regarding the integer constraints. In this study, the widely used expected improvement (EI) function serves as the acquisition function (Zhan and Xing 2020).

4.2.1. Integer-restricted EI function

Let $f^* = \min\{f_1, \dots, f_m\}$. The fundamental concept of EI is to sample a new solution $\mathbf{x}$ to maximize the expected improvement over the current best sample (Kleijnen et al. 2012), that is,

$$\max_{\mathbf{x}} \mathbb{E}[\max(f^* - f(\mathbf{x}), 0)] \quad (33)$$

Given the predictive distribution (32), the optimization problem (33) is equivalent to.

$$\max_{\mathbf{x}} (f^* - \mu_f(\mathbf{x})) \Phi\left(\frac{f^* - \mu_f(\mathbf{x})}{\sigma_f(\mathbf{x})}\right) + \sigma_f(\mathbf{x}) \phi\left(\frac{f^* - \mu_f(\mathbf{x})}{\sigma_f(\mathbf{x})}\right) \quad (34)$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ are the cumulative distribution function (CDF) and probability density function (PDF) of the standard normal distribution, respectively.

While using EI functions to suggest new samples for the land-air collaboration composite transportation networks, constraints of the original network design problem should be incorporated, such that feasible solutions can be sampled and evaluated. Therefore, an integer-restricted EI function is formulated as follows:

4.2.2. A branch-and-bound method

$$\max_{\mathbf{x}} (f^* - \mu_f(\mathbf{x})) \Phi\left(\frac{f^* - \mu_f(\mathbf{x})}{\sigma_f(\mathbf{x})}\right) + \sigma_f(\mathbf{x}) \phi\left(\frac{f^* - \mu_f(\mathbf{x})}{\sigma_f(\mathbf{x})}\right) \quad (35)$$

subject to.

$$c_l \leq c \leq c_u \quad (36)$$

$$z_i \in \{0, 1\}, i = 1, \dots, P \quad (37)$$

The integer-restricted EI function is solved in the branch-and-bound framework. The branch-and-bound (B&B) method employs a tree diagram, consisting of nodes and branches, to structure the partitioning of solutions. Specifically, the algorithm begins by solving a relaxed EI function (all binary constraints are relaxed). This results in the first node, which contains the relaxed solution. The feasibility and objective function value of the relaxed solution are evaluated to determine: (i) the upper bound of the node, (ii) the lower bound of the problem, and (iii) the further branching of the problem. Subsequently, for each branch of the problem, a relaxed EI function—with certain specific relaxed binary constraints—is solved, yielding a relaxed solution. This process continues until a feasible solution is reached, which possesses the highest upper bound value among all terminal nodes. Note that the EI function is highly nonlinear and nonconvex. A challenge of adopting the B&B algorithm is how to derive the upper bound of the EI function. This study adopts the method proposed by Jones et al. (1998) to estimate upper bounds. The B&B algorithm for solving the integer-restricted EI function is summarized in Algorithm 1.

Algorithm 1. Branch-and-bound algorithm to solve the integer-restricted EI function

Input: The Gaussian process surrogate model of $f(\cdot)$; the minimal objective function value f^* of all samples.

Output: The new sample $\mathbf{x}_{m+1}$.

```

1: Let  $\mathcal{S}$  denote the search space derived by relaxing all binary constraints*. Initialize  $L = \{\mathcal{S}\}$ ,  $f_L^* = -\infty$ .
2: While  $L \neq \emptyset$  do
3:     Select a search space  $S$  from  $L$ ;
4:     Optimize the EI function (35) in  $S$ . Let  $\mathbf{x}_S^*$  denote the optimal solution.
5:     If  $\mathbf{x}_S^*$  is a feasible solution do
6:         If  $f(\mathbf{x}_S^*) > f_L^*$  do
7:             Update  $\mathbf{x}^* = \mathbf{x}_S^*$ ,  $f_L^* = f(\mathbf{x}_S^*)$ .
8:         End if
9:     Else
10:        If  $f(\mathbf{x}_S^*) > f_L^*$  do
11:            Select a variable  $x_i$  for branching**.
12:            Partition  $S$  into  $S1 = \{\mathbf{x} | \mathbf{x} \in S, x_i = 1\}$ , and  $S2 = \{\mathbf{x} | \mathbf{x} \in S, x_i = 0\}$ . Let  $L = L \cup \{S1, S2\}$ .
13:        End if
14:    End if
15:     $L = L \setminus \{S\}$ 
16: End while
17: Return  $\mathbf{x}^*$ .
```

Notes: *The binary constraints are replaced by $0 \leq p_i \leq 1$, $i = 1, \dots, P$. **This study randomly selects a binary variable, which has a fractional part in $\mathbf{x}_S^*$, as the branching variable.

Subsequently, the MI-BO algorithm is applied to solve the proposed model, and the comprehensive procedure is summarized in Algorithm 2.

Algorithm 2 Mixed-integer Bayesian optimization algorithm

Input: The objective function $f(\cdot)$; the number m of initial sample points.

Output: A satisfactory solution $\mathbf{x}^*$.

(continued on next page)

(continued)

Algorithm 2 Mixed-integer Bayesian optimization algorithm**Input:** The objective function $f(\cdot)$; the number m of initial sample points.**Output:** A satisfactory solution $\mathbf{x}^*$.

- 1: Randomly generate m initial sample points, $X_m = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$. Evaluate the samples by solving the lower-level problem using the IGP method. Denote the objective function values as $F_m = (f_1, \dots, f_m)$.
 - 2: **While** the stopping criteria are not met **do**
 - 3: Construct the Gaussian process surrogate model of $f(\cdot)$ according to (32).
-
- 1: Randomly generate m initial sample points, $X_m = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$. Evaluate the samples by solving the lower-level problem using the IGP method. Denote the objective function values as $F_m = (f_1, \dots, f_m)$.
 - 2: **While** the stopping criteria are not met **do**
 - 3: Construct the Gaussian process surrogate model of $f(\cdot)$ according to (32).
-
- 4: Use Algorithm 1 to solve P1 and obtain a new solution $\mathbf{x}_{m+1}$.
 - 5: Based on the newly determined solution $\mathbf{x}_{m+1}$, use a path-based IGP algorithm to solve the lower-level problem (i.e., LAC-NE model).
 - 6: Solve the lower-level problem and obtain the objective function value $f_{m+1} = f(\mathbf{x}_{m+1})$ associated with $\mathbf{x}_{m+1}$.
 - 7: Update $X_{m+1} = X_m \cup \{\mathbf{x}_{m+1}\}$, $F_{m+1} = (f_1, \dots, f_m, f_{m+1})$, $m = m + 1$.
 - 8: **End while**
 - 9: Let $\mathbf{x}^* = \arg\min_{i=1, \dots, m} f_i$. **Return** $\mathbf{x}^*$
-

Remark 1: To facilitate the construction of a composite network, upon establishing the decisions made by the upper-level model (i.e., vertiport locations and capacities), each vertiport is connected to the node in the road network that is nearest to it. Furthermore, connections are established between any two vertiports. This process transforms the lower-level problem into a unique land-air collaboration network equilibrium (LAC-NE) problem. It is crucial to note that the LAC-NE problem differs from the traditional user equilibrium (UE) problem due to variations in the impedance function characterizations of various links. Consequently, the algorithm conventionally used to solve the UE problem can be adapted to address the LAC-NE model. In this context, we opt for the advanced path-based IGP algorithm (Xie et al. 2018) to effectively tackle the LAC-NE model.

Remark 2: There is randomness in Algorithm 2, which stems from the generating of initial samples. There are two ways to mitigate this randomness. The first approach involves i) running multiple replications of the algorithm to generate various network design schemes and ii) evaluating each one to select the scheme with the best performance. This method is commonly used in practical

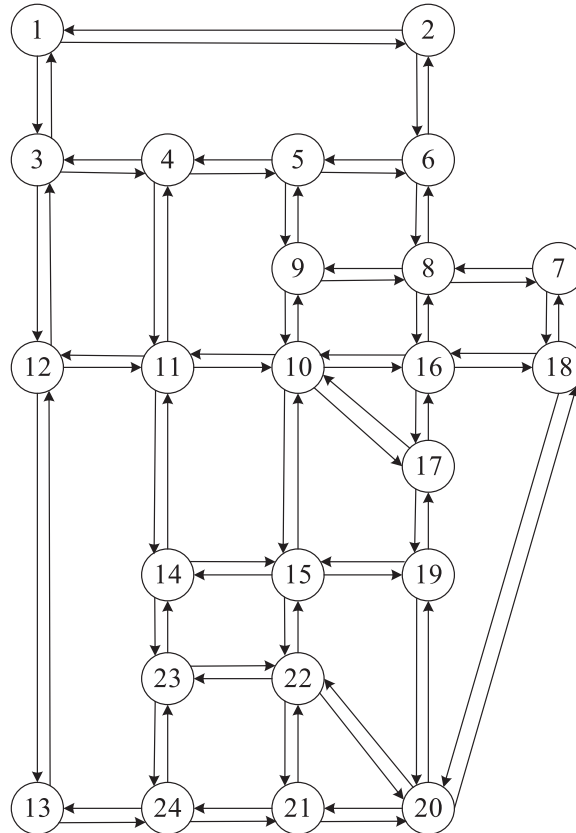


Fig. 5. The road transportation network.

applications. The second is to increase the computational budget by selecting a larger number of samples. As the number of samples increases, the randomness in results decreases.

5. Numerical examples

In this section, we provide several numerical examples to validate our proposed models and methods. In Section 5.1, experiments are carried out on the Sioux Falls network, demonstrating that the land-air collaborative network design model can effectively alleviate congestion. Section 5.2 features experiments on the large-scale Anaheim network. The objective here is to showcase the effectiveness of the proposed methods within real-world networks and to confirm the efficiency of the MI-BO. Details about the experimental networks are accessible at the website: <https://github.com/bstabler/TransportationNetworks/tree/master>. All experiments are performed on a Windows 10 64-bit computer equipped with an AMD 4800H 2.9 GHz CPU and 16G RAM. The proposed algorithm is implemented in C++ using Visual Studio 2022.

5.1. Experiments on the Sioux-Falls network

The first experiment is built from the Sioux Falls network. The tested road network, as depicted in Fig. 5, consists of 24 nodes and 76 links, and there are a total of 576 OD pairs. The link lengths are determined based on the geographical coordinates (longitude and latitude) of the respective nodes. Other parameters incorporated into the optimization model are based on reasonable values from existing studies. For instance, the free-flow speed on both the road links and the evacuation links is set at 60 km/h (Wang et al. 2020b). The horizontal cruising speed for land-air amphibious vehicles is established at 300 km/h, while the average speed for vertical take-off and landing is set to 50 km/h (Bacchini et al. 2021; Li and Liu 2022). The safe spacing for land-air amphibious vehicles during the flight phase is maintained at 10 m. Notably, we assume that there are eight alternative vertiports available for the authority to choose. Furthermore, the solution precision of the low-level problem (i.e., the LAC-NE model) is evaluated using the relative gap (RG) (Xie et al. 2018; Zhang et al. 2023a), with the IGP algorithm concluding when the RG is below 10^{-12} . The MI-BO algorithm is employed to solve the bi-level programming model. The stopping criterion for MI-BO is defined as reaching a predetermined computational budget, which in this case corresponds to 75 iterations.

As detailed in Section 3, we primarily focus on addressing the composite transportation network design problem with land-air coordinated operations, where the central challenge is determining the optimal number of vertiports and their capacity limits. The significance of these two parameters lies in two key aspects: (i) the number and location of vertiports shape the topology of the land-air

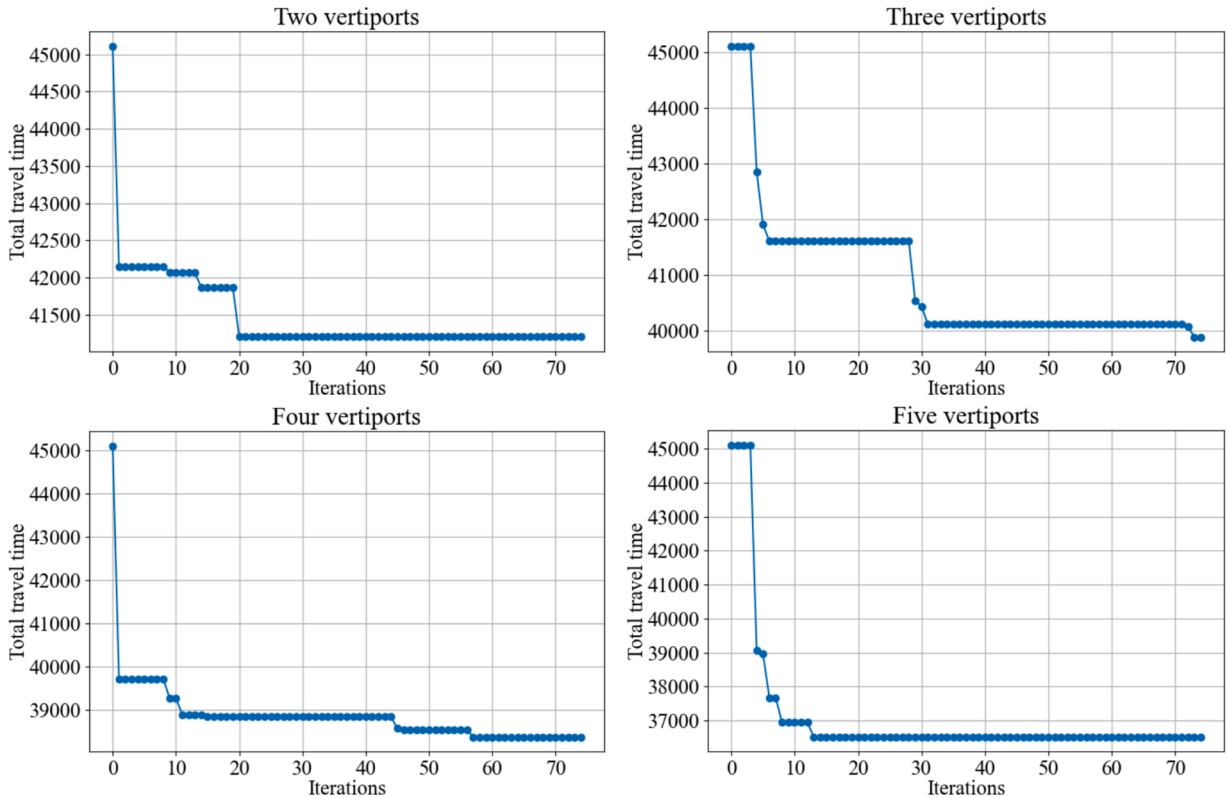


Fig. 6. Convergence performance of the MI-BO algorithm.

cooperative network, which directly influences the set of alternative routes available to travelers; (ii) the capacity of vertiports impacts the impedance value of low-altitude travel, influencing travelers' mode choices, which in turn affects the distribution of travel demand and the total travel time of the transportation system. Based on this analysis, we conducted a comprehensive sensitivity analysis of these two key parameters.

Fig. 6 evaluates, for different numbers of vertiports, the performances of the proposed method as a function of the iteration number. The horizontal axis displays the iteration number, and the vertical axis denotes the objective function value (i.e., total travel time) of the best point identified so far by the method. In all four cases, the proposed method could identify satisfactory solutions within the first few iterations. This demonstrates the efficiency of the proposed method. In addition, Fig. 6 indicates that with the increase in the number of vertiports, there is an observed decline in the total travel time. This highlights the necessity of constructing LAC composite networks.

Subsequently, Table 2 provides a comprehensive overview of network performance, comparing scenarios with and without the integration of UAM. It becomes evident from this table that the inclusion of UAM in the design of the composite transportation network leads to a significant reduction in total travel time. Notably, the optimal capacity of vertiports exhibits variability, as both vertiport capacity and location are determined by the regulatory authorities. Moreover, with an increase in the number of established vertiports, there is a proportional increase in the sharing rate of land-air collaboration, leading to an additional decrease in the total travel time.

To illustrate, the strategic construction of five vertiports to facilitate the LAC operations yields a noteworthy 19.1 % reduction in the total travel time of the transportation system when compared to the scenario without considering the concept of UAM. In simple terms, when government authorities propose alternative plans for vertiport construction, the model developed in this paper can be used to determine the optimal locations and capacities of the vertiports. Additionally, based on the transportation system's operational efficiency and various cities' development plans for the low-altitude economy, the land-air collaborative network design model can help government departments identify the most suitable number of vertiports, while also analyzing the impact of UAM on travel demand distribution and traffic congestion from a macro perspective. This would lead to more informed and effective planning and design of the land-air cooperative transportation network.

In what follows, we focus on investigating the distribution of the ratio of link flow to link capacity (i.e., V/C) across diverse network designs, as shown in Fig. 7. Insightful, the advent of eVTOL vehicles offers a potential avenue for alleviating urban congestion issues through the strategic development of a land-air collaboration composite transportation network.

Notably, compared to the conventional urban transportation network that neglects consideration for the UAM, effectively determining the location and capacity of vertiport can yield a substantial increase in the number of links with V/C values below 1.0, signifying a state of low congestion. In contrast, the count of links experiencing high congestion (i.e., V/C greater than 1.0) can be reduced to a certain extent. Furthermore, with the establishment of more vertiports, such as five in number, the positive impact on alleviating urban congestion becomes more pronounced. However, it is imperative to comprehensively evaluate this positive effect against the economic costs associated with vertiport construction. Determining an equilibrium point requires a nuanced consideration, a top that warrants further exploration in subsequent studies.

5.2. Experiments on the Anaheim network

The Anaheim network, as shown in Fig. 8, has 416 nodes and 914 links, with a total number of 1406 OD pairs. In this experiment, we assume that there are 16 alternative vertiports available for the authority to choose. We aim to select eight vertiports out of 16. Parameter settings on the network design problem and methods are the same as those in Section 5.1.

The MI-BO method is applied to solve the network design problem. For comparison, its performance is evaluated against the GA method, which is widely used in network design studies. Both methods are run 10 times using the same initial solution. Fig. 9 shows the convergence curves for each method (MI-BO in blue, GA in red). The x-axis represents the number of traffic assignment model invocations, the most time-consuming aspect of solving network design problems, especially in large-scale networks. The y-axis shows the total travel time of the traffic system. The solid line represents the mean value across 10 experiments, with the variance also depicted. The results demonstrate that the proposed MI-BO method achieves both higher solution accuracy and efficiency. Starting from the initial solution, MI-BO reduced the total travel time to 1,250,000 s, taking approximately 1/8 of the time required by GA. Moreover, MI-BO shows stable performance across the 10 experiments, indicating its effectiveness in consistently identifying satisfactory solutions.

We tested the impact of capacity values on total network travel time, based on the optimal network design solutions. As depicted in Fig. 10, it shows the total travel time of the network under different capacity values. As illustrated, within the selected vertiports, the variation in capacity values has a slight effect on total travel time. This suggests that in practical implementation; by selecting

Table 2
Network performance with and without incorporating the UAM.

Indicators	Without UAM	Two vertiports	Three vertiports	Four vertiports	Five vertiports
Vertiport capacity	-	10000	10000	9511	10000
Auto share	100 %	96.4 %	94.9 %	92.9 %	90.4 %
LAC share	0.0 %	3.6 %	5.1 %	7.1 %	9.6 %
Total travel time	45114.9	41199.7	39891.0	38371.0	36524.9

Notes: LAC denotes the land-air collaboration.

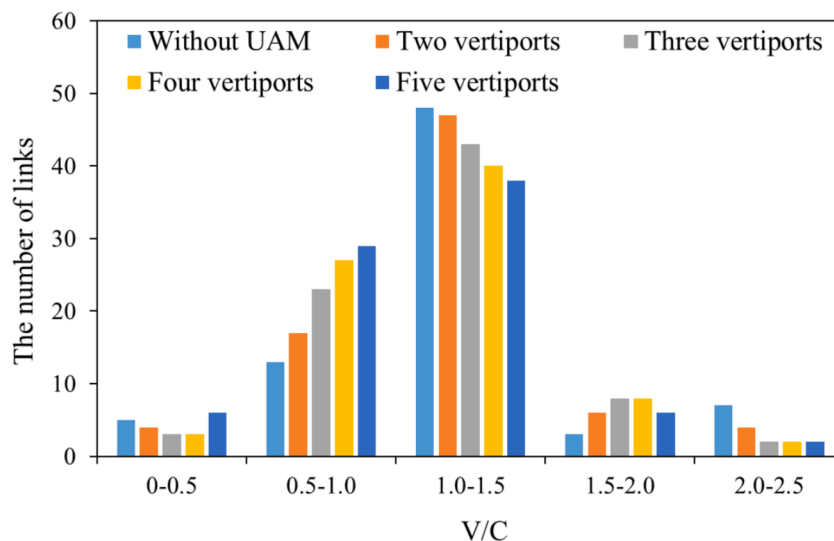


Fig. 7. The distribution of V/C in different network design schemes.

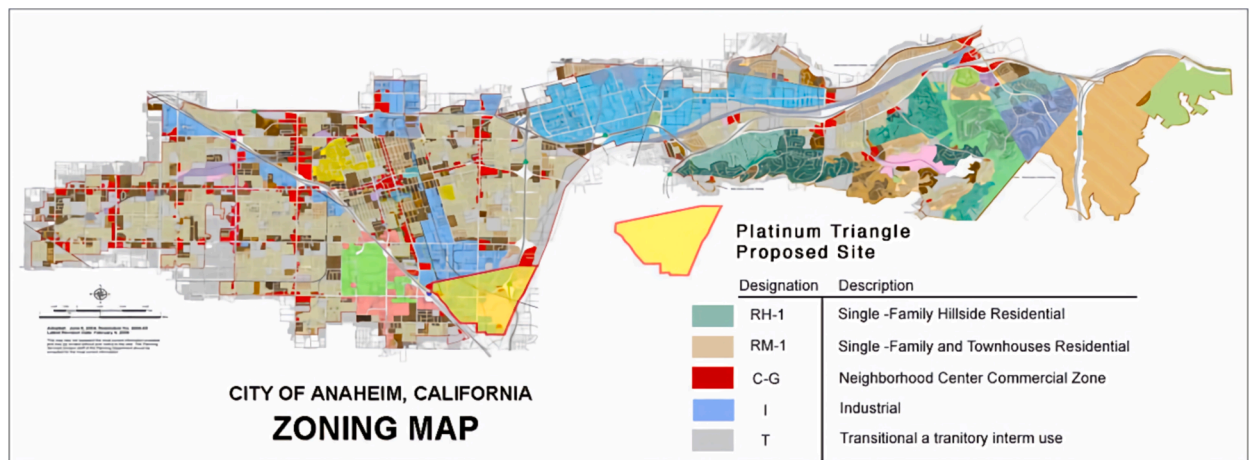


Fig. 8. The layout of the Anaheim network.

appropriate vertiports, lower capacity values can be used to meet network travel demands.

We then explored the impact of UAM on ground traffic congestion under varying levels of traffic demand. Specifically, we scaled the original traffic demand by factors of 0.4, 0.8, 1, 1.2, and 1.6. For each demand level, we solved the CTNDP to generate a corresponding network design scheme. We then evaluated network performance before and after incorporating UAM. The results are presented in Table 3, which summarizes the demand distribution, flow distribution over the ground network, and total travel time. For the flow distribution, we calculated the percentage of links with varying V/C ratios.

Table 3 reveals the following insights: i) As traffic demand increases, the proportion of demand served by UAM slightly decreases. ii) In terms of flow distribution, integrating UAM into the transportation network significantly reduces the proportion of congested links on the original road network, particularly by decreasing the percentage of mildly congested links within the urban network. Observing the percentage of congested links across different demand levels, we find that UAM's congestion mitigation effect becomes more pronounced under high-traffic demand conditions. These findings indicate that UAM can effectively alleviate road traffic congestion. iii) This conclusion is further supported by the total travel time: as traffic demand grows, the reduction in total travel time due to UAM implementation becomes more substantial.

6. Conclusions

This paper developed an innovative framework for modeling the land-air coordinated operations, where the emerging concept of UAM is integrated with conventional travel modes. Focusing on the land-air collaboration composite transport networks, we proposed

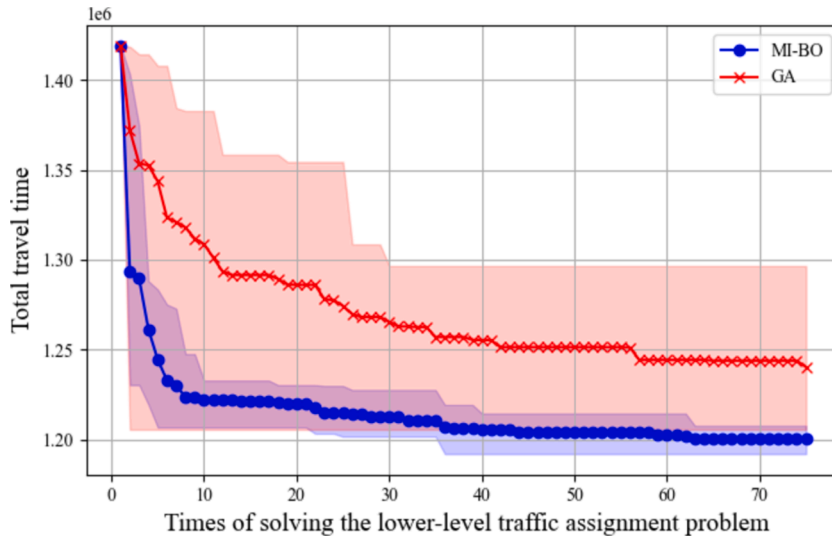


Fig. 9. Convergence curves of each method.

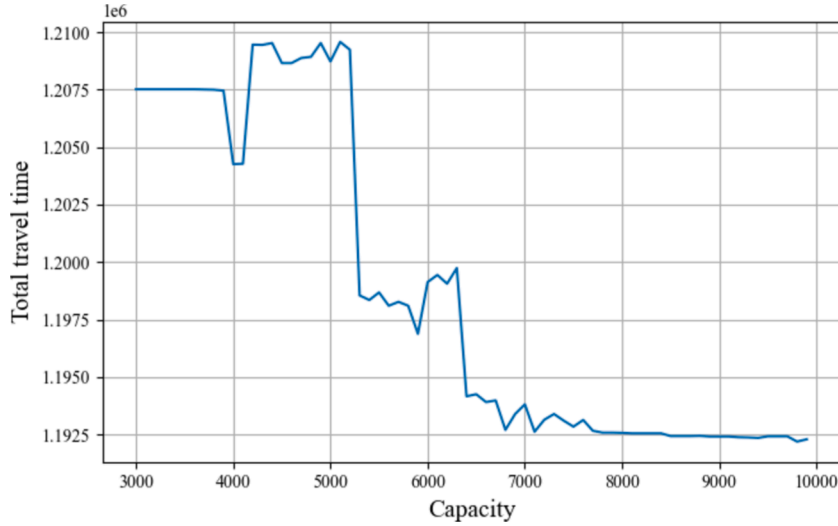


Fig. 10. Sensitivity analysis of capacity values.

a bi-level modeling framework for the addressed topic. The objective of the upper level is to determine optimal vertiport locations and capacities, minimizing the total travel time of the transportation system. The lower-level problem, capturing commuters' route choice behavior in the composite transport network, is solved using a mathematical programming model. A tailored MI-BO method, incorporating an efficient IGP algorithm for the LAC-NE problem, was developed to solve the proposed model. Numerical experiments validated the effectiveness of the analysis framework and the solution method. With the emergence of eVTOL vehicles, the proposed model can provide decision support for authorities and the formulation of relevant management policies. In summary, the proposed model exhibits a higher degree of foresight in developing land-air coordinated operations, offering broad potential applications as a planning tool.

Notably, the proposed optimization model serves as a foundational framework and represents an initial exploration of the UAM concept within urban transportation networks. Future extensions could include: (i) considering additional travel modes (e.g., metro, transit, and bike) when investigating the LAC-CTNDP. (ii) addressing the stochastic nature of travel demands or travel times to enhance the robustness and reliability of the composite transportation network. (iii) incorporating realistic factors related to eVTOL vehicles in the behavioral component of the lower-level problem to better characterize travelers' behavior when choosing eVTOL vehicles for daily commuting. (iv) regulating the LAC-CTNDP based on the time-dependent transportation services.

Table 3
Network Performance with and without UAM across Traffic Demand Levels.

Demand level		0.4	0.8	1.0	1.2	1.6
Anaheim network without UAM						
Demand distribution (%)	Auto	100	100	100	100	100
	LAC	0	0	0	0	0
Flow distribution (%)	0–0.5	95.95	78.75	75.36	71.85	64.73
	0.5–1	3.94	18.84	18.18	17.31	17.96
	1–1.5	0.11	2.19	6.13	9.75	13.14
	1.5–2	0.00	0.22	0.22	0.88	3.50
	2–	0.00	0.00	0.11	0.22	0.66
Total travel time		501641	1061180	1418920	1868570	3263760
Anaheim network with UAM						
Demand distribution (%)	Auto	97.05	97.29	97.35	97.42	97.56
	LAC	2.95	2.71	2.65	2.58	2.44
Flow distribution (%)	0–0.5	97.83	84.47	80.13	75.46	67.21
	0.5–1	1.95	13.79	15.85	17.59	20.20
	1–1.5	0.22	1.52	3.47	5.97	8.90
	1.5–2	0.00	0.22	0.54	0.87	3.04
	2–	0.00	0.00	0.00	0.11	0.65
Total travel time		428474	900598	1194270	1567260	2757140

CRedit authorship contribution statement

Honggang Zhang: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Jinbiao Huo:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Churong Chen:** Visualization, Validation, Investigation, Formal analysis. **Zhiyuan Liu:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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